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Moderating Indonesian Code-Mixed Hate Speech: IndoBERTweet and Tiered Guardrail Trade-Offs

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Abstract

Automated moderation of Indonesian social media is complicated by code-mixing among Indonesian, colloquial English, Javanese, and Sundanese, and by evaluations whose sample sizes are inflated by augmented variants of the same text. This study examines how accurately and how safely a fine-tuned IndoBERTweet classifier can support a multi-tiered guardrail with human-in-the-loop review. We curated 199,360 texts from an Indonesian Twitter corpus and two code-mixed datasets, partitioned them with GroupShuffleSplit on source-text groups, and evaluated a frozen checkpoint in FP32 with 1,000 grouped bootstrap resamples. On non-augmented test texts (N = 3,907) the model obtained an F1-score of 0.8647 (95% CI 0.8518–0.8763) and an AUROC of 0.9391; on the full test set (40,934 texts from 4,100 groups) the F1-score was 0.8773. It exceeded a TF-IDF logistic regression baseline by 0.0527 F1 on original texts and 0.0167 on all texts, and outperformed zero-shot Qwen2.5-1.5B on 400 independent groups (accuracy 0.895 versus 0.613–0.650). Softmax scores were overconfident (ECE 0.0964): 95.9% fell outside 0.10–0.90, and the intermediate tiers held only 2.1% of traffic with near-chance precision. Raising the blocking threshold from 0.50 to 0.90 reduced wrongly blocked benign texts from 10.06% to 8.13% while 444 hateful texts moved to masking, whereas temperature scaling (T = 2.8091) cut ECE to 0.0348 and benign blocking to 2.41%. Guardrails therefore reallocate errors rather than remove them, and calibration is a prerequisite for meaningful tiers.

Keywords: code-mixed hate speech; human-in-the-loop; IndoBERTweet; probability calibration; tiered guardrail

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1. Introduction

Hate speech on social media harms targeted individuals and erodes public discourse, making automated screening essential at scales that human moderators cannot handle alone. For Indonesian platforms, foundational efforts established multi-label Twitter datasets [1], while subsequent reviews have documented ongoing domain challenges [2]. A central difficulty is that hateful and merely offensive expressions share substantial vocabulary, causing classifiers that rely on surface lexical cues to confuse benign profanity with genuine hate speech [3]. In operational moderation, every false positive silences a legitimate user, whereas every false negative leaves harmful content visible.

Indonesian online communication further complicates moderation through pervasive code-mixing among standard Indonesian, colloquial English, and regional languages such as Javanese and Sundanese, alongside non-standard abbreviations. While recent studies have explored word-level domain transfer for code-mixed hate speech [4], multilingual BERT adaptation for Telugu code-mixing [5], cross-lingual pre-training [6], and implicit coded hate detection [7], Indonesian regional code-mixing remains under-served. Prior work spans contextual embeddings [8], recurrent networks over static vectors [9], explainable Twitter classifiers [10], and pretrained transformers including BERT [11] , domain-specific IndoBERTweet [12], HateBERT [13], and context-synergized networks [14]. Furthermore, functional benchmarks such as HateCheck [15], HateXplain [16], bias mitigation studies [17], and cross-dataset evaluations [18] show that aggregate accuracy often masks brittle lexical shortcuts. Recently, large language models (LLMs) have been evaluated for regional Spanish detection [19] and counter-speech generation [20], while Indonesian monitoring systems [21], multilingual human-in-the-loop (HITL) pipelines [22], and class-imbalance metric analyses [23] underscore that classifier scores are only one part of a safe moderation workflow.

Three methodological gaps motivate this study. First, evaluations on automatically augmented code-mixed corpora rarely separate independent source texts from derived variants, inflating nominal test sizes and narrowing row-level confidence intervals. Second, multi-tiered guardrails are frequently proposed without auditing whether underlying softmax probabilities are calibrated, how blocking thresholds trade wrongful blocks against residual exposure, or how queue precision degrades under realistic low-prevalence traffic. Third, comparisons against zero-shot LLMs often rely on small samples and uncalibrated default thresholds. We address three research questions: how accurately fine-tuned IndoBERTweet classifies four-language code-mixed text under group-preserving evaluation compared with classical and zero-shot LLM baselines (RQ1); how probability calibration and threshold policies govern operational guardrail trade-offs across prevalence regimes (RQ2); and how errors distribute across dialects, lexical triggers, text lengths, and normalization stages (RQ3). Our contributions provide a leakage-controlled dual-reporting evaluation, an empirical calibration and guardrail policy audit, a checksum-verified shared-core HITL application with verified GPU-to-CPU numerical parity, and a transparent accounting of dataset and modelling limitations.

2. Methods

2.1 Dataset Curation and Group-Preserving Partitioning Protocol

The corpus integrates the Indonesian Twitter dataset of Ibrohim and Budi [1] (where each tweet forms one independent group) with the Indonesian–Javanese and Indonesian–Sundanese subsets of the code-mixed hate speech dataset (aiatums/codemixed-id-hate-speech, CC BY 4.0). In the code-mixed subsets, each record contains one original text and two augmented variants bound to a single group identifier (group_id) generated from the MD5 hash of the normalized original text. Expanding all records yielded 283,887 raw rows. Removing empty strings, filtering 954 texts with contradictory cross-row labels (11,503 rows removed, including 11,441 Sundanese original rows), and deduplicating exact normalized texts (272,008 to 199,360 rows; 26.7% reduction) produced a cleaned corpus of 199,360 rows (111,761 non-hate and 87,599 hate speech; 43.94% positive). Because contradiction filtering used full-corpus labels prior to splitting, the resulting partitions may be cleaner than raw platform traffic.

Partitioning used GroupShuffleSplit on group_id (seed 42; 20% test, 10% validation from remaining groups) to guarantee that an original text and its augmented variants never cross partitions, yielding 142,647 training, 15,779 validation, and 40,934 internal test rows (Table 1). Strict assertions verified zero overlap of group_id or normalized text across splits. An official held-out test set was constructed from the released Javanese and Sundanese test splits (original texts only) and filtered against empty, contradictory, duplicate, and internally overlapping groups, reducing 6,000 raw rows to 1,995 texts (996 Javanese, 999 Sundanese). Because 37,027 of the 40,934 internal test rows (90.5%) are augmented variants derived from 1,546 code-mixed groups (~24 rows per group) while only 3,907 rows are unaugmented originals, we report both original-only and full-test metrics and resample strictly at the group level.

Table 1. Dataset composition and group-preserving partition summary (rows; unique groups in parentheses; hate speech prevalence in %)

Source domain	Train	Validation	Internal test	Official held-out test
Indonesian Twitter	9,382 (9,382); 42.02%	1,046 (1,046); 42.93%	2,554 (2,554); 42.99%	not available
Indonesian–Javanese	66,684 (2,678); 62.39%	7,118 (285); 55.55%	19,757 (793); 61.71%	996 (996); 60.54%
Indonesian–Sundanese	66,581 (2,698); 26.05%	7,615 (309); 22.68%	18,623 (754); 28.37%	999 (999); 34.53%
Total	142,647 (14,758*); 44.09%	15,779 (1,640*); 38.85%	40,934 (4,100); 45.37%	1,995 (1,995); 47.52%
Original texts only	14,075 (14,075); 42.47%	1,562 (1,562); 41.43%	3,907 (3,907); 43.56%	1,995 (1,995); 47.52%

**Sum of per-domain unique groups; a single group identifier occurs in both code-mixed languages in the test partition (4,101 versus 4,100), so train and validation totals may be overstated by a similar amount.*

2.2 Four-Language Lexical Normalization and Shared Core Architecture

Input texts undergo deterministic normalization: case folding, regular-expression removal of hexadecimal escapes, URLs, user mentions, and crawler placeholders (rt, user, url), and token-level mapping via a merged dictionary while preserving boundary punctuation. The dictionary combines the 15,195-entry Indonesian colloquial lexicon [1] with a curated four-language lexicon covering English slang, Javanese, and Sundanese variants (e.g., mapping stfu to shut the fuck up, asu and anjay to anjing, pekok to bodoh, and balegug to belegug). A separate 47-term lexicon is reserved exclusively for masking sensitive words in user-facing displays and never alters model inputs.

To prevent code drift between training and deployment, all core operations—text normalization, display masking, threshold mapping, policy wording, and an autocast-disabled FP32 inference routine—are

implemented once in a shared module (hateguard_core.py, SHA-256 e07ad4c7…). Both the training pipeline and the deployment application verify this module's SHA-256 digest and a four-sentence preprocessing parity hash (0ebe0f02…) before execution, halting immediately on any mismatch.

2.3 Model Fine-Tuning, Classical Baseline, and Zero-Shot LLM Protocol

The primary classifier is IndoBERTweet base uncased [12], a bidirectional transformer [11] implemented via Hugging Face Transformers and Datasets [24], [25]. Fine-tuning used a maximum length of 128 tokens, batch size 16, learning rate 2×10^{-5}, weight decay 0.01, two epochs, seed 42, and mixed precision on a Tesla T4 GPU, selecting the best checkpoint by validation F1. Evaluation was executed in verification mode on the frozen checkpoint (model.safetensors, SHA-256 3ec22197…). Provenance checks confirmed exact matches for weight checksums, configuration parameters, and the 40,934 internal test and 1,995 held-out texts; however, because the legacy manifest omitted a standalone training configuration block and exhibited different split-fingerprint strings, training and validation set membership relies on configuration parity rather than independent hash proof.

For a classical baseline, we trained a logistic regression classifier on identical splits using combined word (1–2 gram, 300,000 max features) and character (2–5 char_wb, 400,000 max features) sublinear TF-IDF vectors, selecting regularization C = 2.0 on validation (F1 = 0.8435). For a zero-shot LLM comparator, we evaluated Qwen2.5-1.5B-Instruct on raw texts across 400 independent original-text groups (200 per class: 259 Twitter, 82 Javanese, 59 Sundanese) and a legacy 50-group sample. Alongside generative label parsing, we extracted a parser-free score defined as the first-token logit difference between the HATE_SPEECH and NON_HATE_SPEECH labels $s_i = \ell_i^{\mathrm{HATE}} - \ell_i^{\mathrm{NON}}$, evaluating it at threshold 0 and at a threshold (3.8984) calibrated on 200 disjoint validation groups.

2.4 Multi-Tiered Guardrail Policy, Probability Calibration, and HITL System

For each text i, the classifier produces FP32 logits $(z_{i,0}, z_{i,1})$ and hate speech softmax probability $p_i = exp(z_{i,1}) / \left(exp(z_{i,0}) + exp(z_{i,1})\right)$. The deployed guardrail maps p_i into four operational action levels using thresholds $\tau_1 = 0.50$, $\tau_2 = 0.75$, and $\tau_3 = 0.90$ according to Equation (1) :

$$\mathrm{level}(p_i) = \begin{cases} \text{SAFE}, & p_i < \tau_1 \\ \text{WEAK}, & \tau_1 \le p_i < \tau_2 \\ \text{MODERATE}, & \tau_2 \le p_i < \tau_3 \\ \text{STRONG}, & p_i \ge \tau_3 \end{cases} \quad (1)$$

In Equation (1), SAFE comments are published normally, WEAK comments are published with lexical masking, MODERATE comments are published with masking and a spoiler warning, and STRONG comments are blocked and escalated to the HITL moderator queue. We compared six policies on the test set: binary blocking at 0.50 (A) and 0.90 (B), mask-only at 0.50 (C), three-tier masking at 0.50 and blocking at 0.90 (D), the deployed four-tier policy (E), and four-tier routing on temperature-scaled probabilities (F). Post-hoc temperature scaling divides logits by $T > 0$ fitted by minimising validation negative log-likelihood (NLL), yielding calibrated probability $p_i^{(T)} = \sigma\left(\left(z_{i,1} - z_{i,0}\right)/T\right)$ without altering binary predictions at 0.50. Calibration is quantified via 10-bin expected calibration error (ECE) in Equation (2), where B_b denotes bin b, yi the true label, and N the total rows, alongside Brier score :

$$\mathrm{ECE} = \sum_{b=1}^{10} \frac{1}{N} \left| \sum_{i \in B_b} (y_i - p_i) \right| \quad (2)$$

To assess deployment under lower hate speech prevalence π, we projected queue precision assuming fixed class-conditional test block rates $r_1 = P(block|hate)$ and $r_0 = P(block|non-hate)$ via Equation (3) :

$$Prec_b lock(\pi) = \frac{\pi r_1}{\pi r_1 + (1-\pi) r_0} \quad (3)$$

The Gradio HITL application enforces startup checksum and parity gates (tolerance 10^{-3} across 200 stratified texts), lazily loads Qwen2.5-1.5B-Instruct only for STRONG items to draft advisory reasoning and polite rewrites (falling back to lexical masking on error), and logs atomic moderator decisions, SHA-256 digests, and truncation flags (>128 tokens) into an SQLite audit database with configurable text redaction and retention.

2.5 Statistical Evaluation and Reproducibility Protocol

All evaluations use pure FP32 inference. Uncertainty is quantified via 95% percentile confidence intervals from 1,000 grouped bootstrap resamples (500 for AUROC), resampling unique group_id clusters with replacement. For any metric f derived from group confusion counts c_g across G groups with multinomial resample weights $w_g^{(b)}$ $Multinomial(G, 1/G)$, the bootstrap replicate $M^{(b)}$ and paired system difference $\Delta^{(b)}$ between models A and B follow Equation (4) :

$$M^{(b)} = f\left(\sum_{g=1}^{G} w_g^{(b)} c_g\right), \qquad \Delta^{(b)} = f\left(\sum_{g=1}^{G} w_g^{(b)} c_g^A\right) - f\left(\sum_{g=1}^{G} w_g^{(b)} c_g^B\right). \quad (4)$$

We report accuracy, precision, recall, F1, prevalence-robust balanced accuracy, and AUROC [23], plus exact McNemar tests on independent paired LLM samples. Validity checks include a TF-IDF cosine near-duplicate audit (threshold 0.80, max_df= 1%), a 128-token truncation audit, an inference-time normalization ablation (V0 full, V1 without four-language lexicon, V2 without dictionaries, V3 lowercase only), and a standalone CPU smoke test in an isolated Python process.

3. Results and Discussions

3.1 Classification Performance Across Data Partitions and Baselines

Table 2 reports primary classification performance with 95% grouped bootstrap intervals. On the 3,907 original test texts (1 text per group), fine-tuned IndoBERTweet achieved 0.8815 accuracy [0.8707, 0.8915], 0.8647 F1 [0.8518, 0.8763], 0.8801 balanced accuracy, and 0.9391 AUROC [0.9310, 0.9458]. Across all 40,934 test rows (4,100 groups), row-weighted accuracy was 0.8888 and F1 was 0.8773 (AUROC 0.9460), while equal group weighting gave 0.8940 accuracy and 0.8794 F1. Although the 37,027 augmented rows (accuracy 0.8895) did not heavily skew point estimates, they inflate nominal sample size tenfold over the 4,100 independent groups. The confusion matrices in Figure1 show a 10.06% FPR and 12.40% FNR on all test rows (10.93% and 13.04% on originals). On the official held-out split (1,995 Javanese and Sundanese originals), the model reached 0.8607 accuracy, 0.8487 F1, and 0.9172 AUROC.

Compared with the TF-IDF logistic regression baseline, IndoBERTweet achieved statistically significant paired F1 gains on every subset except Sundanese originals (Table 2), led by natural non-augmented text: +0.0527 [+0.0394, +0.0660] on original texts, +0.0638 on Indonesian Twitter, and +0.0515 on the held-out split, versus +0.0167 on all test rows. However, the baseline matched IndoBERTweet in pooled test AUROC (0.9471 vs. 0.9460), suggesting that fine-tuning primarily improves separation at the 0.50 operating threshold and on natural expressions where surface n-grams falter [3], [12], [13]. Performance was lower on Sundanese originals (0.6640 F1, 0.8047 balanced accuracy, 0.8862 AUROC at 20.50% hate prevalence after removing 11,441 contradictory rows) than on Javanese originals (0.8800 F1 at 61.74% prevalence). Validity audits confirmed that only 3.76% of original test texts (147/3,907) had cosine similarity >= 0.80 to training texts (excluding them shifts accuracy only from 0.8815 to 0.8777), only 0.30% (6/1,995) were flagged in the held-out split, and 0% of test texts exceeded 128 tokens (maximum 118 tokens).

Table 2. Primary classification performance: fine-tuned IndoBERTweet versus TF-IDF + logistic regression baseline (hate class positive; ↑ higher is better; 95% CI from 1,000 grouped bootstrap resamples, AUROC 500; ΔF1 is a paired grouped bootstrap difference).

Subset (rows / groups)	Model	Accuracy [95% CI]	Precision	Recall	F1 [95% CI]	Bal. acc.	AUROC [95% CI]	ΔF1 IndoBERTweet − TF-IDF [95% CI]
Original texts only (3,907 / 3,907)	IndoBERTweet	0.8815 [0.8707, 0.8915]	0.8600	0.8696	0.8647 [0.8518, 0.8763]	0.8801	0.9391 [0.9310, 0.9458]	+0.0527 [+0.0394, +0.0660]
	TF-IDF + LR	0.8395 [0.8277, 0.8503]	0.8291	0.7955	0.8120 [0.7974, 0.8257]	0.8345	0.9215 [0.9127, 0.9293]	—
All test, row-weighted (40,934 / 4,100)	IndoBERTweet	0.8888 [0.8793, 0.8968]	0.8785	0.8760	0.8773 [0.8655, 0.8876]	0.8877	0.9460 [0.9388, 0.9521]	+0.0167 [+0.0099, +0.0232]
	TF-IDF + LR	0.8747 [0.8647, 0.8835]	0.8690	0.8523	0.8606 [0.8477, 0.8718]	0.8728	0.9471 [0.9405, 0.9522]	—
Javanese code-mixed, augmented (18,965 / 793)	IndoBERTweet	0.8971 [0.8847, 0.9092]	0.9190	0.9138	0.9164 [0.9054, 0.9271]	0.8920	0.9549 [0.9452, 0.9629]	+0.0107 [+0.0034, +0.0182]
	TF-IDF + LR	0.8835 [0.8706, 0.8959]	0.9053	0.9060	0.9057 [0.8941, 0.9173]	0.8766	0.9530 [0.9441, 0.9606]	—
Javanese code-mixed, original (792 / 792)	IndoBERTweet	0.8523 [0.8283, 0.8775]	0.8827	0.8773	0.8800 [0.8580, 0.9010]	0.8446	0.9171 [0.8957, 0.9384]	+0.0305 [+0.0074, +0.0538]

Subset (rows / groups)	Model	Accuracy [95% CI]	Precision	Recall	F1 [95% CI]	Bal. acc.	AUROC [95% CI]	ΔF1 IndoBERTweet − TF-IDF [95% CI]
	TF-IDF + LR	0.8220 [0.7955, 0.8485]	0.8884	0.8139	0.8495 [0.8251, 0.8721]	0.8244	0.9049 [0.8836, 0.9257]	—
Sundanese code-mixed, augmented (18,062 / 754)	IndoBERTweet	0.8816 [0.8688, 0.8941]	0.7935	0.7925	0.7930 [0.7660, 0.8177]	0.8549	0.9265 [0.9162, 0.9371]	+0.0214 [+0.0057, +0.0376]
	TF-IDF + LR	0.8730 [0.8600, 0.8866]	0.7951	0.7494	0.7715 [0.7440, 0.7968]	0.8360	0.9283 [0.9179, 0.9384]	—
Sundanese code-mixed, original (561 / 561)	IndoBERTweet	0.8485 [0.8199, 0.8788]	0.6087	0.7304	0.6640 [0.5961, 0.7319]	0.8047	0.8862 [0.8513, 0.9175]	+0.0442 [-0.0150, +0.1119]
	TF-IDF + LR	0.8360 [0.8039, 0.8681]	0.5906	0.6522	0.6198 [0.5446, 0.6923]	0.7678	0.8834 [0.8527, 0.9140]	—
Indonesian Twitter (2,554 / 2,554)	IndoBERTweet	0.8978 [0.8857, 0.9096]	0.8815	0.8807	0.8811 [0.8664, 0.8951]	0.8957	0.9540 [0.9459, 0.9614]	+0.0638 [+0.0476, +0.0800]
	TF-IDF + LR	0.8457 [0.8324, 0.8591]	0.8327	0.8024	0.8173 [0.8000, 0.8339]	0.8404	0.9275 [0.9180, 0.9376]	—
Official held-out test (1,995 / 1,995)	IndoBERTweet	0.8607 [0.8446, 0.8747]	0.8764	0.8228	0.8487 [0.8302, 0.8656]	0.8589	0.9172 [0.9057, 0.9294]	+0.0515 [+0.0328, +0.0710]
	TF-IDF + LR	0.8206 [0.8020, 0.8371]	0.8606	0.7426	0.7973 [0.7749, 0.8162]	0.8169	0.9115 [0.8985, 0.9234]	—

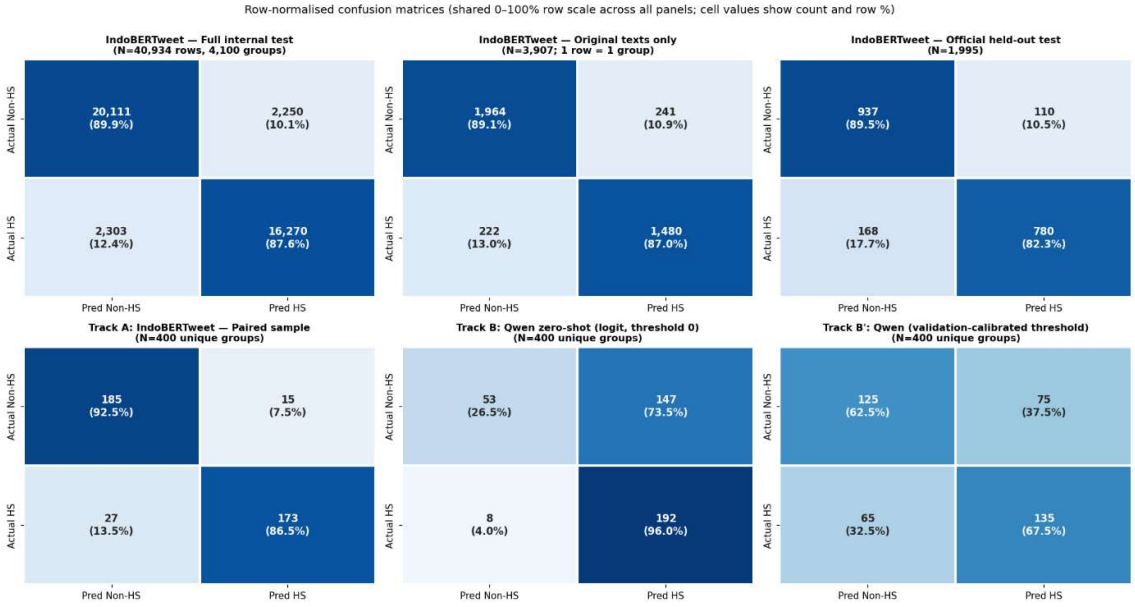


Figure 1. Row-normalised confusion matrices for six evaluation conditions (shared 0–100% scale; counts in cells) across internal test (N = 40,934 rows; 4,100 groups), original-only test (N = 3,907), official held-out split (N = 1,995), and paired 400-group samples for IndoBERTweet and Qwen2.5-1.5B (↑ higher diagonal is better; seed = 42).

3.2 Zero-Shot LLM Comparison and Threshold Calibration

Table 3 compares fine-tuned IndoBERTweet against zero-shot Qwen2.5-1.5B-Instruct across 400 independent original-text groups. IndoBERTweet achieved 0.8950 accuracy [0.8650, 0.9250], 0.8918 F1, and 0.9545 AUROC. Under the default threshold, Qwen obtained 0.6125 accuracy and 0.7124 F1, but labelled 84.8% of all inputs as hateful. Because the generative parsing route (with zero parse failures) and the first-token logit route at threshold 0 yielded identical predictions, this deficit reflects decision bias rather than formatting errors. McNemar's exact test on the 169 discordant pairs (141 correct only for IndoBERTweet vs. 28 for Qwen) confirmed a highly significant difference ($p < 0.001$; paired accuracy gain +0.2825 [+0.2225, +0.3401]).

Calibrating Qwen's logit threshold (3.8984) on 200 disjoint validation groups (validation balanced accuracy 0.745, AUROC 0.7884) balanced its test predictions (52.5% positive) and raised test accuracy to 0.6500 [0.6025, 0.7000], but F1 fell to 0.6585 and IndoBERTweet maintained a wide lead (+0.2450 accuracy, +0.2332 F1, $p < 0.001$; AUROC 0.9545 vs. 0.7250). The legacy 50-group sample replicated this result (0.9200 vs. 0.5000 uncalibrated and 0.6000 calibrated, $p < 0.001$). In line with studies exploring LLMs on regional language varieties [19], this finding is bounded by a single 1.5B model, one zero-shot prompt, and unnormalized text.

Table 3. Paired evaluation against zero-shot Qwen2.5-1.5B-Instruct on independent groups (N = 400 primary original-text groups; hate class positive; ↑ higher is better; 95% CI from 1,000 grouped bootstrap resamples, AUROC 500; McNemar exact test; seed = 42).

Sample	System	Accuracy [95% CI]	F1 [95% CI]	Bal. acc.	AUROC [95% CI]	Predicted HS	McNemar b / c; p	ΔAccuracy (IndoBERTweet − Qwen) [95% CI]	ΔF1 [95% CI]
N = 400 groups	IndoBERTweet fine-tuned	0.8950 [0.8650, 0.9250]	0.8918 [0.8579, 0.9239]	0.8950	0.9545 [0.9326, 0.9721]	47.0%	—	—	—
N = 400 groups	Qwen2.5-1.5B, generative label (penalized) = logit threshold 0*	0.6125 [0.5675, 0.6625]	0.7124 [0.6704, 0.7557]	0.6125	0.7250 [0.6785, 0.7746]	84.8%	141 / 28; p<0.001	+0.2825 [+0.2225, +0.3401]	+0.1793 [+0.1288, +0.2280]
N = 400 groups	Qwen2.5-1.5B, logit threshold 3.8984 (validation-calibrated)	0.6500 [0.6025, 0.7000]	0.6585 [0.6042, 0.7137]	0.6500	0.7250 [0.6785, 0.7746]	52.5%	122 / 24; p<0.001	+0.2450 [+0.1875, +0.3000]	+0.2332 [+0.1688, +0.2946]

**The generative route (unparseable answers counted as errors) and the logit route at threshold 0 produced identical predictions; AUROC is computed from the logit score. N = 400: 259 Twitter, 82 Javanese, 59 Sundanese originals.*

3.3 Guardrail Policy Ablation, Probability Calibration, and Prevalence Projection

The classifier's raw softmax scores are severely overconfident, with test ECE of 0.0964 [0.0890, 0.1056], Brier score of 0.1022, NLL of 0.5348, and 95.9% of scores falling below 0.10 or at or above 0.90 (Figure 2). This is consistent with validation loss rising from 0.455 to 0.567 across epochs 1 and 2 while validation F1 rose. Consequently, the intermediate WEAK (405 rows, 0.99%) and MODERATE (472 rows, 1.15%) tiers hold only 2.14% of traffic and yield near-chance empirical hate rates of 45.68% [39.75%, 51.71%] and 54.87% [49.22%, 60.42%], compared with 10.27% in SAFE (22,414 rows) and 89.70% in STRONG (17,643 rows).

Table 4 shows that tiered guardrails redistribute rather than eliminate errors. Raising the blocking threshold from 0.50 (Policy A) to 0.90 with intermediate masking (Policies D and E) reduces wrongly blocked benign texts from 2,250 (10.06%) to 1,817 (8.13%, saving 433 benign texts) and lowers HITL load from 45.24% to 43.10%, but shifts 444 hateful texts to masked publication and leaves 12.40% (2,303 hateful texts) unmoderated in SAFE (Figure 3). Policies D and E yield identical exposure metrics because the 0.75 boundary only alters warning labels. Applying validation-fitted temperature scaling ($T = 2.8091$; Policy F) cut test ECE to 0.0348 [0.0301, 0.0396], Brier score to 0.0874, benign blocking to 2.41% (540 texts), and HITL load to 30.04% while raising block precision to 95.61% and restoring monotonic tier hate rates (55.13% WEAK, 77.18% MODERATE) [22], while routing 4,513 hateful texts to lexical masking. Under prior-shift projections, if hate prevalence dropped from 45.37% to 10%, 5%, or 1%, deployed queue precision would fall from 89.70% to 53.81% (15.83% load), 35.56% (11.98% load), and 9.58% (8.90% load; 80.4 wrongful blocks vs. 1.24 unmasked hateful texts per 1,000 comments at 1% prevalence).

Table 4. Guardrail policy ablation on the internal test set (N = 40,934 rows from 4,100 groups; 95% CI from 1,000 grouped bootstrap resamples; temperature T = 2.8091 fitted on validation only; percentages).

Policy	HITL load	Benign blocked [95% CI]	HS published unmoderated	HS masked only (n)	Block precision [95% CI]
A. Binary block, τ = 0.50	45.24	10.06 [9.05, 11.30] (n = 2,250)	12.40 (n = 2,303)	0	87.85 [86.15, 89.29]
B. Binary block, τ = 0.90	43.10	8.13 [7.25, 9.21] (n = 1,817)	14.79 (n = 2,747)	0	89.70 [88.15, 91.04]
C. Mask-only, τ = 0.50 (no HITL)	0.00	0.00 [0.00, 0.00] (n = 0)	12.40 (n = 2,303)	16,270	—
D. Three-tier (mask 0.50, block 0.90)	43.10	8.13 [7.25, 9.21] (n = 1,817)	12.40 (n = 2,303)	444	89.70 [88.15, 91.04]
E. Four-tier, deployed (0.50/0.75/0.90)	43.10	8.13 [7.25, 9.21] (n = 1,817)	12.40 (n = 2,303)	444	89.70 [88.15, 91.04]
F. Four-tier on temperature-scaled scores	30.04	2.41 [1.99, 2.92] (n = 540)	12.40 (n = 2,303)	4,513	95.61 [94.59, 96.42]

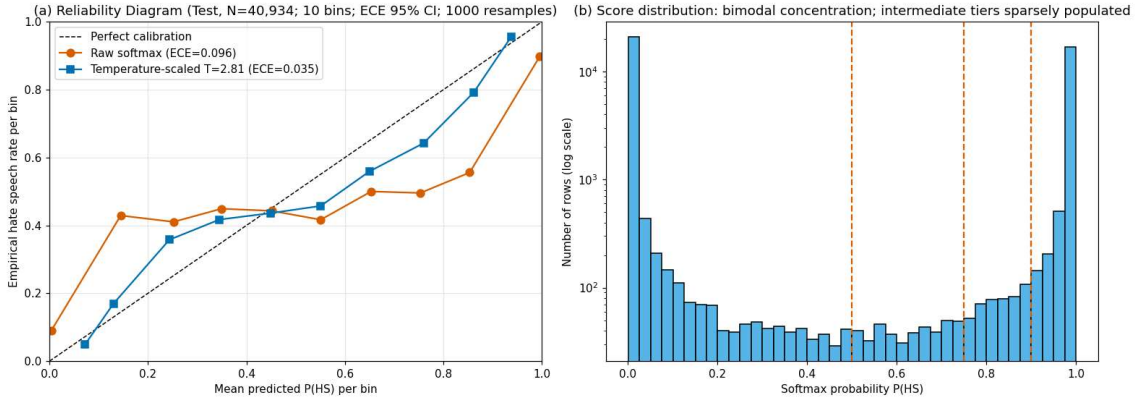


Figure 2. Probability calibration on internal test (N = 40,934 rows; 4,100 groups): (a) 10-bin reliability diagram for raw softmax (ECE 0.0964) and temperature-scaled scores (T = 2.8091, ECE 0.0348; ↓ lower is better; 1,000 resamples); (b) log-scale score histogram with thresholds 0.50, 0.75, and 0.90 (seed = 42).

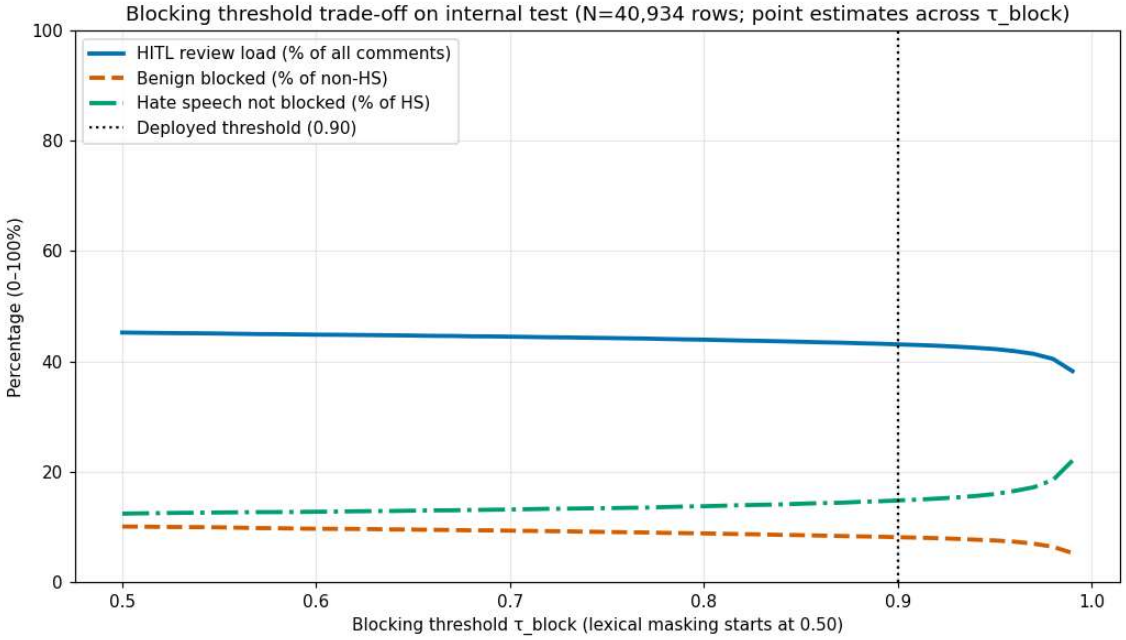


Figure 3. Trade-off across blocking thresholds τ_block in [0.50, 0.99] on internal test (N = 40,934 rows; masking at 0.50) for HITL load, benign blocked rate, and unblocked hate speech rate (↓ lower is better; seed = 42).

3.4 Normalization Ablation and Multi-Axis Error Analysis

Inference-time normalization ablation on the frozen model (full table in repository) showed small but consistent F1 reductions across all test rows when removing the four-language lexicon (V1: ΔF1 = +0.0047 [+0.0025, +0.0071]), all dictionaries (V2: +0.0054), or regex cleaning (V3: +0.0054). On original texts, removing the four-language lexicon alone spanned zero (+0.0023 [−0.0010, +0.0057]), whereas removing all dictionaries or cleaning reduced F1 by +0.0105 and +0.0141, concentrated in Indonesian Twitter (V3: +0.0175 [+0.0062, +0.0285]). Because the model was not retrained per variant, these differences measure train–test mismatch rather than training-time contribution.

Lexical error analysis (Table 5) reveals that benign false positives cluster on regional and colloquial profanity. Against a 10.06% baseline benign FPR, benign texts containing Sundanese belegug reached 46.51% FPR (4.62× risk ratio), followed by jancuk (21.43%, 2.13×), asu (17.13%, 1.70×), cok (14.42%), anjay (14.29%), and anjir (12.23%), whereas identity terms cina (9.38%) and kafir (12.50%) remained near baseline with wide intervals. High recall on hateful occurrences of these slurs (84.8%–99.1%) is consistent with lexical shortcut reliance [3], [15]. Across text lengths, accuracy was stable for 1–40 words (0.8709–0.9061) but fell to 0.8461 (16.67% FPR) above 40 words. Across domains, the model over-flagged Javanese originals (18.81% FPR vs. 12.27% FNR) and under-detected Sundanese originals (26.96% FNR vs. 12.11% FPR).

Table 5. Lexicon-associated false-positive risk analysis on the internal test set (N = 40,934 rows from 4,100 groups; overall benign FPR = 10.06% [9.05, 11.30]; 95% CI from 1,000 grouped bootstrap resamples)

Term (normalized to)	Rows / groups	Benign rows	False positives	FPR [95% CI]	Risk ratio	Recall on hateful rows with term
belegug (belegug)	407 / 152	86	40	46.51% [34.28, 57.50]	4.62×	89.41%
jancuk (jancok)	913 / 302	252	54	21.43% [12.72, 31.25]	2.13×	91.23%
asu (anjing)	2,578 / 444	607	104	17.13% [11.40, 24.36]	1.70×	87.72%
cok (jancok)	1,117 / 300	416	60	14.42% [9.01, 20.49]	1.43×	93.72%
anjay (anjing)	34 / 29	28	4	14.29% [0.00, 32.26]	1.42×	100.00%
kafir (kafir)	125 / 65	16	2	12.50% [0.00, 31.26]	1.24×	99.08%
anjir (anjing)	1,532 / 569	1,006	123	12.23% [8.87, 15.82]	1.22×	84.79%
idiot (bodoh)	59 / 40	9	1	11.11% [0.00, 40.00]	1.10×	96.00%
cina (cina)	80 / 68	32	3	9.38% [0.00, 27.28]	0.93×	95.83%
kapir (kafir)	22 / 12	0	0	not defined	—	100.00%

3.5 Discussion, Operational Implications, and Study Limitations

Addressing RQ1, fine-tuned IndoBERTweet substantially outperforms lexical and zero-shot baselines on natural texts (0.8647 F1 on originals; +0.0527 over TF-IDF), whereas pooled augmented evaluation (+0.0167) obscures both this advantage and the weaker Sundanese original performance (0.6640 F1). Addressing RQ2, uncalibrated softmax scores (95.9% extreme) leave intermediate tiers nearly empty; guardrails trade wrongful blocks against masked hateful exposure, and validation temperature scaling (T = 2.8091) is required to cut benign blocking (8.13% to 2.41%) and HITL load (43.10% to 30.04%) while leaving SAFE false negatives (12.40%) unchanged. Addressing RQ3, errors reflect opposite dialectal biases (high Javanese FPR, high Sundanese FNR) and lexical shortcuts on slurs such as belegug.

Operationally, platforms must calibrate probabilities and tune thresholds to target prevalence rather than benchmark priors, treating small LLM rewrites strictly as unverified moderator aids [20], [21], [22]. Our shared FP32 core and SHA-256 gates prevented deployment drift: whereas the original Trainer-based predictions differed from FP32 by up to 0.0188 in probability (1 label flip and 5 tier flips across 40,934 rows), the CPU

application and standalone smoke test matched GPU FP32 references within 1.27×10^{-5} and 1.55×10^{-5} with zero tier flips. Key limitations bound our findings: augmented labels were not human-verified; 11,503 contradictory rows and 26.7% duplicates were removed pre-split; evaluation used a single frozen checkpoint (one seed, no hyperparameter tuning, and unverified train/validation split fingerprints in the legacy manifest); the held-out set shares the source corpus; normalization ablation was inference-only; and no human-moderator study was conducted.

4. Conclusions

This study evaluated a fine-tuned IndoBERTweet classifier within a four-tier human-in-the-loop guardrail for Indonesian, English-slang, Javanese, and Sundanese code-mixed social media under a group-preserving protocol. On 3,907 non-augmented original test texts, the model achieved 0.8647 F1 and 0.9391 AUROC (0.8773 F1 and 0.9460 AUROC across all 40,934 test rows from 4,100 groups), outperforming a TF-IDF logistic regression baseline by +0.0527 F1 on original texts (+0.0167 on all rows) and surpassing zero-shot Qwen2.5-1.5B by 0.245 to 0.283 in accuracy across 400 independent groups ($p < 0.001$). Nevertheless, performance varied across dialects (0.6640 F1 on Sundanese vs. 0.8800 on Javanese originals), and benign false positives clustered on colloquial slurs such as belegug (46.51% FPR). Crucially, raw softmax probabilities were severely overconfident (ECE 0.0964, with 95.9% of scores outside 0.10–0.90), leaving intermediate guardrail tiers nearly empty (2.14% traffic) at near-chance precision. Raising the blocking threshold from 0.50 to 0.90 reduced benign blocking from 10.06% to 8.13% only by shifting 444 hateful texts to lexical masking, whereas validation-fitted temperature scaling ($T = 2.8091$) cut ECE to 0.0348, benign blocking to 2.41%, and moderator review load from 43.10% to 30.04% while leaving the 12.40% false-negative rate in the SAFE tier unchanged. Prior-shift projections further showed that deploying benchmark thresholds under low real-world hate prevalence (1%–10%) would fill the review queue predominantly with benign comments. Guardrails therefore reallocate rather than eliminate moderation errors, making probability calibration and group-level evaluation mandatory before deployment. Future work should incorporate multi-seed retraining, human-audited dialectal corpora, cross-platform streams, and live moderator trials.

Declarations

AI Usage Statement

The authors used large language model assistants (Claude, ChatGPT, and Gemini) to assist in refactoring the experiment and application notebooks, auditing the statistical evaluation protocol, and drafting and editing the manuscript text. All code executions, dataset curations, numerical verifications against the frozen evaluation outputs, and final scientific conclusions were conducted and verified by the authors.

Author Contributions

M.R.S.H.: conceptualization, methodology, software implementation, formal analysis, validation, writing – original draft, writing – review and editing; F.L.: data curation, investigation, visualization, writing – review and editing; C.J.S.: investigation, validation, resource preparation, writing – review and editing; V.C.M.: supervision, validation, project administration, funding acquisition, writing – review and editing.

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Conflict of Interest Statement

The authors declare no conflict of interest.

Ethical Clearance

This study used only publicly released text datasets and involved no human participants or animal subjects; therefore no ethical approval number was required. The data contain real hateful and offensive content, which is displayed in masked form in this article and its released outputs, and the moderation prototype stores user text locally with options for redaction, backup control, and retention.

Data and Code Availability

The Indonesian Twitter hate speech data are available from the repository of Ibrohim and Budi [1], and the code-mixed data from the Hugging Face dataset aiatums/codemixed-id-hate-speech (CC BY 4.0, commit 19a03eba); the encoder and the language model are available as indolem/indobertweet-base-uncased and Qwen/Qwen2.5-1.5B-Instruct (both Apache-2.0). The notebooks, shared core module, aggregated results (RESULTS_SINGLE_SOURCE.json), and checksums are available at https://github.com/michaelrespati/HateGuard_NLP.git. Model weights are not stored in the repository and are identified by SHA-256 3ec22197cb5248be993782d490cc080efb9f2d0f341ba7884f1b7d99ff31e0df; the shared module has SHA-256 e07ad4c7524b1ed3e30fa0fa5a91a9163c40ab4a0547663afc15ef7f861a91b6. Moderation logs are not released because they may contain sensitive text.

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